MINISTÉRIO DA CIÊNCIA, TECNOLOGIA E INOVAÇÃO
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

aa/bb/cc/dd-TDI

ASSESSING VULNERABILITY THROUGH GRAPH CENTRALITY AND EFFICIENCY ANALYSIS: A GEOSPATIAL ANALYSIS OF THE TAMANDUATEÍ RIVER BASIN

Emmanuel Inioluwa Babajide

Qualifying Examination for the Doctorate in the Postgraduate Program in Applied Computing, supervised by Dr. Leonardo Bacelar Lima Santos and Dr. Rogerio Galante Negri.

URL do documento original:
<http://urlib.net/xx/yy>

INPE
São José dos Campos
2026

PUBLICADO POR:

Instituto Nacional de Pesquisas Espaciais - INPE
Coordenação de Ensino, Pesquisa e Extensão (COEPE)
Divisão de Biblioteca (DIBIB)
CEP 12.227-010
São José dos Campos - SP - Brasil
Tel.:(012) 3208-6923/7348
E-mail: pubtc@inpe.br

CONSELHO DE EDITORAÇÃO E PRESERVAÇÃO DA PRODUÇÃO INTELECTUAL DO INPE - CEPPII (PORTARIA Nº 176/2018/SEI-INPE):

Presidente:

Dra. Marley Cavalcante de Lima Moscati - Coordenação-Geral de Ciências da Terra (CGCT)

Membros:

Dra. Ieda Del Arco Sanches - Conselho de Pós-Graduação (CPG)

Dr. Evandro Marconi Rocco - Coordenação-Geral de Engenharia, Tecnologia e Ciência Espaciais (CGCE)

Dr. Rafael Duarte Coelho dos Santos - Coordenação-Geral de Infraestrutura e Pesquisas Aplicadas (CGIP)

Simone Angélica Del Ducca Barbedo - Divisão de Biblioteca (DIBIB)

BIBLIOTECA DIGITAL:

Dr. Gerald Jean Francis Banon

Clayton Martins Pereira - Divisão de Biblioteca (DIBIB)

REVISÃO E NORMALIZAÇÃO DOCUMENTÁRIA:

Simone Angélica Del Ducca Barbedo - Divisão de Biblioteca (DIBIB)

André Luis Dias Fernandes - Divisão de Biblioteca (DIBIB)

EDITORAÇÃO ELETRÔNICA:

Ivone Martins - Divisão de Biblioteca (DIBIB)

André Luis Dias Fernandes - Divisão de Biblioteca (DIBIB)

MINISTÉRIO DA CIÊNCIA, TECNOLOGIA E INOVAÇÃO
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

aa/bb/cc/dd-TDI

ASSESSING VULNERABILITY THROUGH GRAPH CENTRALITY AND EFFICIENCY ANALYSIS: A GEOSPATIAL ANALYSIS OF THE TAMANDUATEÍ RIVER BASIN

Emmanuel Inioluwa Babajide

Qualifying Examination for the Doctorate in the Postgraduate Program in Applied Computing, supervised by Dr. Leonardo Bacelar Lima Santos and Dr. Rogerio Galante Negri.

URL do documento original:
<http://urlib.net/xx/yy>

INPE
São José dos Campos
2026

Dados Internacionais de Catalogação na Publicação (CIP)

Sobrenome, Nomes.

Cutter Assessing Vulnerability through Graph Centrality and Efficiency Analysis: A Geospatial Analysis of the Tamanduateí River Basin / Nome Completo do Autor1; Nome Completo do Autor2. – São José dos Campos : INPE, 2026.

xxiii + 23 p. ; (aa/bb/cc/dd-TDI)

Dissertação ou Tese (Mestrado ou Doutorado em Nome do Curso) – Instituto Nacional de Pesquisas Espaciais, São José dos Campos, AAAA.

Orientador : José da Silva.

1. Palavra chave. 2. Palavra chave 3. Palavra chave. 4. Palavra chave. 5. Palavra chave I. Título.

CDU 000.000

Esta obra foi licenciada sob uma Licença Creative Commons Atribuição-NãoComercial 3.0 Não Adaptada.

This work is licensed under a Creative Commons Attribution-NonCommercial 3.0 Unported License.

Informar aqui sobre marca registrada (a modificação desta linha deve ser feita no arquivo publicacao.tex).

Informar aqui sobre fontes financiadoras (a modificação desta linha deve ser feita no arquivo publicacao.tex).

ATENÇÃO! A FOLHA DE APROVAÇÃO SERÁ INCLUIDA POSTERIORMENTE.
Mestrado ou Doutorado em Nome do Curso

ASSESSING VULNERABILITY THROUGH GRAPH CENTRALITY AND EFFICIENCY ANALYSIS: A GEOSPATIAL ANALYSIS OF THE TAMANDUATEÍ RIVER BASIN

"You never really understand a system until it breaks".

JEREMY IRONS in
"Margin Call", 2011

To my parents ***Dr. Oluwole Babajide*** *and late mum* ***Prof. Joan Babajide****, strength and guidance continue to inspire every step of this journey.*

AGRADECIMENTOS

To my advisor, Leonardo Santos. To the Postgraduate Program in Applied Computing and to CAPES for the opportunity and scholarship.

RESUMO

Urban transportation networks are increasingly exposed to extreme flooding events that disrupt mobility and infrastructure functionality. This study analyzes the vulnerability of an urban street network using a graph-theoretical framework applied to the Tamanduateí River Watershed (TRW) in São Paulo, Brazil. The street network is represented as a directed graph where intersections correspond to nodes and road segments to edges, with node importance quantified using betweenness centrality and vulnerability defined as the relative reduction in global network efficiency after node removal. Historical flood records (2018–2025) were spatially associated with nearby network nodes, and their centrality and vulnerability distributions compared against randomly sampled non-flood nodes using kernel density estimation (KDE), empirical cumulative distribution functions (ECDF), and non-parametric statistical tests. Results indicate that flood-affected nodes exhibit significantly higher betweenness centrality and vulnerability ($p < 10^{-6}$), with flood-affected nodes showing a median betweenness centrality 156% higher and a median vulnerability 49% greater than non-flood nodes, corresponding to a 0.24% versus 0.16% relative reduction in global network efficiency upon node removal. These findings show that flood events are disproportionately associated with structurally important intersections, amplifying network disruption risk.

Palavras-chave: Graph Theory. Urban Transportation Networks. Betweenness Centrality. Global Efficiency. Network Vulnerability.

AVALIAÇÃO DA VULNERABILIDADE POR MEIO DA CENTRALIDADE DO GRAFO E ANÁLISE DE EFICIÊNCIA: UMA ANÁLISE GEOESPACIAL DA BACIA DO RIO TAMANDUATEÍ

ABSTRACT

As redes de transporte urbano estão cada vez mais expostas a eventos extremos de inundação que interrompem a mobilidade e a funcionalidade da infraestrutura. Este estudo analisa a vulnerabilidade de uma rede viária urbana utilizando uma estrutura teórica de grafos aplicada à Bacia Hidrográfica do Rio Tamanduateí (BHRT) em São Paulo, Brasil. A rede viária é representada como um grafo direcionado onde as interseções correspondem a nós e os segmentos de via a arestas, com a importância dos nós quantificada pela centralidade de intermediação e a vulnerabilidade definida como a redução relativa na eficiência global da rede após a remoção de um nó. Registros históricos de inundações (2018–2025) foram associados espacialmente a nós próximos da rede, e suas distribuições de centralidade e vulnerabilidade foram comparadas com nós não inundados amostrados aleatoriamente utilizando estimativa de densidade de kernel (KDE), funções de distribuição cumulativa empírica (ECDF) e testes estatísticos não paramétricos. Os resultados indicam que os nós afetados por inundações apresentam centralidade de intermediação e vulnerabilidade significativamente maiores ($p < 10^{-6}$), com os nós afetados por inundações mostrando uma centralidade de intermediação mediana 156% maior e uma vulnerabilidade mediana 49% maior do que os nós não afetados por inundações — o que corresponde a uma redução relativa de 0,24% versus 0,16% na eficiência global da rede após a remoção do nó. Essas descobertas mostram que eventos de inundação estão desproporcionalmente associados a interseções estruturalmente importantes, amplificando o risco de interrupção da rede.

Keywords: Teoria dos Grafos. Redes de Transporte Urbano;. Centralidade de Intermediação;. Eficiência Global;. Vulnerabilidade da Rede;.

LISTA DE FIGURAS

Pág.

2.1 Methodological Flowchart 5
2.2 Tamanduatei River Basin 6
2.3 Study Area (a) the street network graph and (b) the historical flood points. 7

3.1 Centrality Distribution Analysis for all Nodes (a) statistical distribution and its (b) geospatial distribution. 13
3.2 Centrality Distribution Analysis for Flood-affected Nodes (a) statistical distribution and its (b) KDE Plot 14
3.3 KDE plot comparing all node categories. 14
3.4 KDE Plot Comparing Node Vulnerabilities 16
3.5 ECDF Plot of (a) Betweenness Centrality and (b) Vulnerability 17

LISTA DE TABELAS

Pág.

2.1 Summary of Network Metrics Used in the Study 9

3.1 Statistical validation and effect size measures for metric distributions (Flood-Affected vs. Random Non-Flood Nodes). 17

LISTA DE ABREVIATURAS E SIGLAS

ECDF	–	Empirical Cumulative Distribution Function
KDE	–	Kernel Density Estimate
KS	–	Kolmogorov–Smirnov
MWU	–	Mann–Whitney U
OSM	–	OpenStreetMap

LISTA DE SÍMBOLOS

n_s	–	Nodes, representing street intersections
E_s	–	Edges, representing road segments
$C_B(n)$	–	Betweenness Centrality of a specific node
σ_{st}	–	Total number of shortest paths between nodes
$\sigma_{st}(n)$	–	Number of shortest paths between s and t that pass through node n
G_E	–	Global Efficiency of the network
N	–	Total number of nodes in the network
d_{ij}	–	Shortest path distance between node i and node j
V_i	–	Vulnerability of an individual node
GE_{original}	–	Global efficiency of the full, undisturbed network
$GE_{\text{without } i}$	–	Global efficiency of the network after node i has been removed

SUMÁRIO

Pág.

1 INTRODUCTION 1

2 MATERIALS AND METHODS 5

2.1 Study Area 5

2.2 Data Sources 6

2.3 Street Network Graph 7

2.4 Node Classification 7

2.5 Network metric Calculation 8

2.6 Analytical Techniques for Comparison and Statistical Validation 10

3 RESULT AND DISCUSSIONS 13

3.1 Betweenness Centrality 13

3.2 Vulnerability 15

3.3 Empirical Cumulative Distribution Function (ECDF) 16

3.4 Statistical Validation 17

4 FINAL REMARKS AND FUTURE WORK 19

REFERÊNCIAS BIBLIOGRÁFICAS 21

1 INTRODUCTION

Urban transportation systems play a fundamental role in sustaining economic activities, social mobility, and access to essential services in modern cities (AHERN, 2011). However, these systems are increasingly susceptible to disruptions caused by extreme weather events, particularly urban flooding (MERZ et al., 2010). Flood events may result in the temporary or permanent interruption of road segments, significantly reducing network connectivity and limiting accessibility across the urban space (BOCCALETTI et al., 2006; ZHAO et al., 2022; BI et al., 2025). As climate-related hazards become more frequent and intense, understanding the structural vulnerability of transportation networks has become an important topic in infrastructure planning and disaster risk management (MEEROW; NEWELL, 2019).

From a mathematical perspective, transportation systems can be modeled as complex networks in which intersections are represented as nodes and road segments as edges (PORTA et al., 2006). This abstraction allows the application of graph theory and network science to analyze structural properties such as connectivity, robustness, and efficiency. Complex network analysis has been widely used to investigate infrastructure systems due to its ability to capture topological characteristics and functional relationships within large-scale networks (BOCCALETTI et al., 2006). In the context of urban transportation, graph-based approaches have been successfully applied to examine structural organization, mobility patterns, and system vulnerability (DERRIBLE; KENNEDY, 2011).

A key concept in network analysis is the identification of structurally important elements that play a critical role in maintaining connectivity. Centrality measures provide quantitative indicators of the relative importance of nodes and edges within a network. Among these measures, betweenness centrality is particularly relevant for transportation networks because it identifies elements that lie on a large number of shortest paths, indicating potential bottlenecks or critical corridors (CRUCITTI et al., 2006). Such metrics have been widely applied in the study of spatial networks to identify infrastructure components whose disruption may cause significant performance degradation.

In addition to local importance measures, global metrics are also necessary to evaluate the overall performance of the system. One widely used indicator is global efficiency, which measures how efficiently information or movement can occur across the network through shortest paths (LATORA; MARCHIORI, 2001). When applied to transportation systems, this measure provides an estimate of how well the

network maintains accessibility between different locations. Reductions in global efficiency may therefore indicate loss of connectivity or increased travel costs due to infrastructure disruptions.

Recent studies have demonstrated that combining centrality measures with global performance indicators provides a robust framework for assessing transportation network vulnerability and resilience (SABERI et al., 2017; KERMANSHAH; DERRIBLE, 2017). By simulating disruptions or removing critical components, it becomes possible to analyze how local failures propagate through the system and affect the overall network structure. Such approaches are particularly relevant in flood-prone urban environments, where localized inundation may generate cascading impacts on mobility and accessibility.

Despite growing research on urban resilience and network analysis, an important gap persists in integrating historical flood data with network criticality metrics. Most existing studies model hypothetical or simulated disruptions rather than empirically observed flood incidents (LV et al., 2025). Consequently, limited attention has been paid to whether past flood events systematically coincide with the most critical nodes in a real-world transportation network. Addressing this gap is vital for developing targeted mitigation strategies that align physical vulnerability with network importance.

To bridge this research gap, the present study employs a graph-based framework to examine the spatial and functional interplay between historical flood events and the structural criticality of urban street networks. Specifically, the study investigates whether flood-affected nodes exhibit higher betweenness centrality and vulnerability compared to non-flood nodes. The analysis integrates visual and statistical methods, including Kernel Density Estimate (KDE) and Empirical Cumulative Distribution Function (ECDF) plots, to explore the distributions of these metrics across node groups. Statistical validation is conducted using non-parametric tests, the Kolmogorov–Smirnov (KS) test and the Mann–Whitney U (MWU) test to determine whether observed differences are statistically significant.

Through this synthesis of spatial, topological, and statistical perspectives, the study contributes to a more nuanced understanding of how environmental hazards intersect with infrastructural criticality. The findings aim to inform evidence-based decision-making among urban planners, engineers, and disaster management authorities, facilitating strategic investments in resilience-building and adaptive design. More broadly, this research underscores the importance of network science in ad-

vancing quantifiable, system-oriented approaches to urban resilience amid escalating climate pressures.

Research Aim:
To quantify the disproportionate impact of historical flood events on the structural criticality and operational efficiency of urban street networks, thereby providing a robust, data-driven foundation for enhanced urban resilience planning and targeted disaster management strategies.

Specific Objectives:

- To model the urban street network using graph theory principles, representing intersections as nodes and road segments as edges, and to compute betweenness centrality for all nodes within the studied area.
- To define and calculate node vulnerability based on the reduction in global efficiency upon individual node removal, establishing a quantitative measure of each node's criticality.
- To conduct a comparative analysis of the distributions of betweenness centrality and vulnerability, utilizing KDE, ECDF plots, and non-parametric statistical tests (Kolmogorov–Smirnov and Mann–Whitney U), between flood-affected nodes and randomly selected non-flood nodes.
- To analyze the correlation between betweenness centrality and vulnerability for both flood-affected and non-flood nodes, and to characterize the network properties (betweenness centrality and vulnerability) of the topmost vulnerable nodes in each group.
- To synthesize the findings to identify systemic vulnerabilities where environmental hazards intersect with critical transportation infrastructure, and to propose actionable insights for urban planning and disaster management to foster resilience.

2 MATERIALS AND METHODS

This chapter addresses the general characteristics of the Tamanduateí River basin, data source, classification, processing, visualization and validation as shown in Figure 2.1.

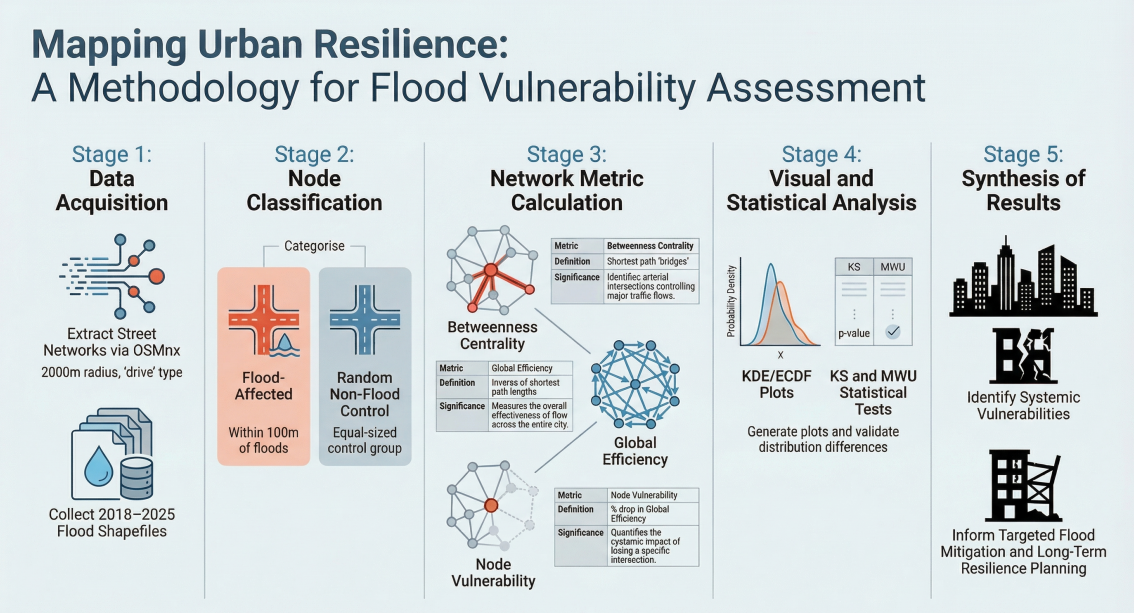

Figura 2.1 - Methodological Flowchart

2.1 Study Area

The Tamanduateí River basin as shown in Figure 2.2 is in the southeastern portion of Sao Paulo Metropolitan Region. It extends across the municipalities of São Paulo, São Caetano do Sul, and Santo André. The basin covers an area of approximately 330 km^2 and drains into the Tietê River, which is the main hydrographic artery of São Paulo. The Tamanduateí has historically played a dual role in urban development: serving as a transport corridor during São Paulo's early industrialization and simultaneously acting as a recurrent source of flooding due to channelization, impermeabilization of soils, and unregulated urban expansion (AB'SÁBER, 2003; MARENGO et al., 2021). As such, the basin represents a natural laboratory for studying the interaction between hydrographic risk and mobility resilience within a megacity context. The primary study area for the analysis is defined as a 2000-meter sample size withing the Tamanduatei basin around the central geographical point

(23.55 °S; 46.62 °W) . This buffer serves to delineate the specific region of the street network and flood data considered for in-depth analysis.

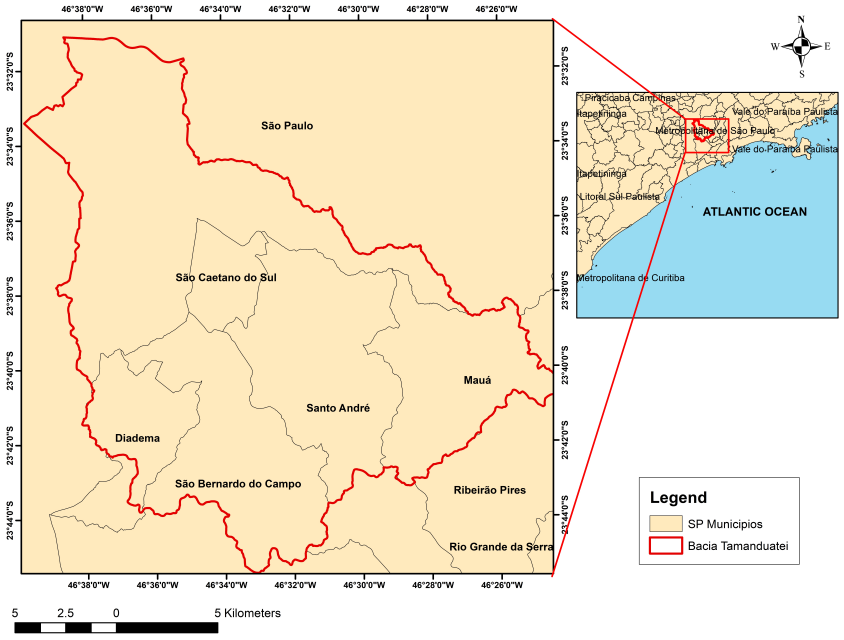

Figura 2.2 - Tamanduatei River Basin

2.2 Data Sources

OpenStreetMap (OSM) Street Networks: The urban street network was extracted directly from OpenStreetMap using the OSMnx Python library (BOEING, 2017). OSM data offered detailed information on the configuration of road segments and intersections within the basin. Through OSMnx, the network was transformed into a graph in which nodes represent intersections and edges correspond to street segments, enabling the computation of graph-theoretical metrics such as degree, centrality, and clustering.

Historical Flood Event Data: Historical flood event data was obtained from two distinct shapefiles, both loaded from Google Drive into Geodata Frames. The shapefiles contain flood incident records from 2018 to April 2025. These shapefiles provide point geometries representing locations where flood events occurred, which were then used to identify 'flood-affected nodes' within the street network.

2.3 Street Network Graph

The street network was modeled as a directed graph $\mathcal{G} = (N, E)$ where nodes $N = \{n_1, \ldots, n_n\}$ represent intersections and edges $E = \{e_{ij} : i, j = 1, \ldots, n\}$ in the directed graph represent street segments, with each edge weighted by its actual road segment length (in metres) as extracted from OpenStreetMap data. (Figure 2.3).

- **Central Coordinates**: The network was generated around the central coordinates within the overall study area. These coordinates represent a significant central location within the study area.
- **Distance Parameter**: A distant parameter of 4 km×4 km from the central coordinate, allowing us to retrieve all streets within a 2000-meter radius from the central point as seen in Figure , defining the geographical extent of our study's street network.
- **Network Type**: The network type was set to 'drive', ensuring that only streets accessible to motor vehicles (e.g., roads, highways, residential streets) were included in the generated graph. This excludes footpaths, cycleways, and other non-vehicular routes, providing a relevant network for analyzing urban traffic vulnerability.

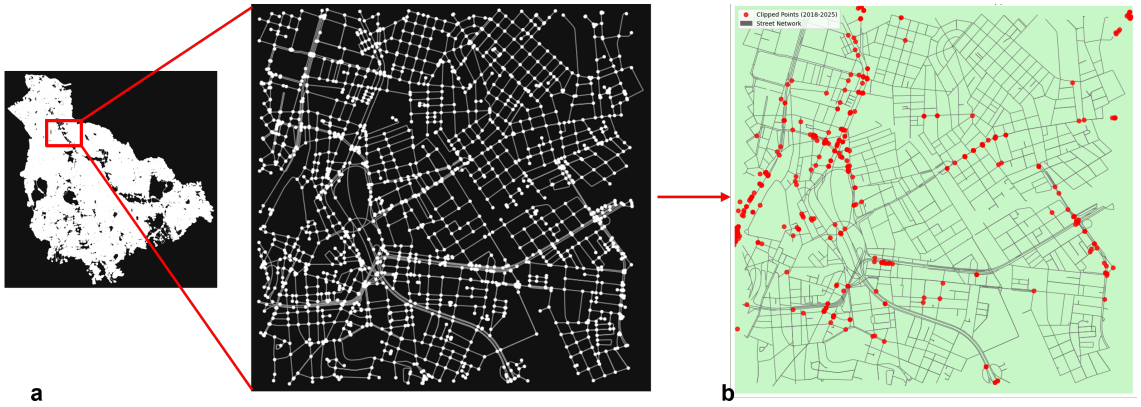

Figura 2.3 - Study Area (a) the street network graph and (b) the historical flood points.

2.4 Node Classification

To effectively analyze the impact of flood events on the urban street network, nodes were classified into two distinct groups: 'Flood-Affected Nodes' and 'Random Non-

Flood Nodes'. This classification allowed for a comparative study of their topological characteristics and vulnerability.

- **Flood-Affected Nodes**: To identify the flood-affected nodes, we identified nodes closest (within 100m) to flood points. This process effectively mapped the geographic flood incidents onto the discrete nodes of the street network, identifying the specific network locations most directly affected by floods. This threshold was selected to account for geolocation uncertainty in historical reports, spatial offsets between reported flood coordinates and mapped intersections, and local street-network spacing.
- **Random Non-Flood Nodes**: For the random non-flood nodes, we first excluded nodes already identified as flood affected nodes, then a random sample of nodes farther than the previously defined radius was carried out. The number of nodes chosen for this random sample was explicitly made to match the total count of the flood affected nodes. This ensures that the two groups being compared have an equal sample size, preventing any bias due to differing group sizes in subsequent statistical analyses.

These random non-flood nodes create a control group that provides a baseline for understanding whether the characteristics observed in flood-affected nodes are specific to their flood status or are merely representative of typical nodes in the network.

2.5 Network metric Calculation

This section details the methodologies employed for calculating key network metrics: betweenness centrality, global efficiency, and node vulnerability, which are crucial for understanding the urban network's structure and resilience.

- **Betweenness Centrality**: Betweenness centrality is a measure of a node's importance in a network, quantifying the number of times a node acts as a bridge along the shortest path between two other nodes (CRUCITTI et al., 2006). In this study, shortest paths were computed using Dijkstra's algorithm with road-segment length (meters) as edge weight, based on OpenStreetMap length attributes. Therefore, nodes with high betweenness centrality are those frequently traversed along minimum-distance travel routes.
- **Global Efficiency Function**: Global efficiency was computed following (LATORA; MARCHIORI, 2001) to quantify the overall efficiency of traffic

flow within the network. For each pair of distinct nodes, the inverse of their shortest path length represents their pairwise efficiency, which is then summed and normalized by the total number of node pairs. A higher global efficiency value indicates a more connected and easily traversable network

- **Node Vulnerability**: Node vulnerability assesses the impact of a single node's removal on the overall global efficiency of the network. This metric helps identify which nodes are critical for maintaining network functionality (CRUCITTI et al., 2006). For disconnected node pairs, where no path exists, distance is considered infinite and their the efficiency contribution for these pairs becomes zero ($1/d_{st}$=0). This approach ensures that the node vulnerability metric (V_i) accurately captures the significant loss of connectivity and accessibility across the urban space when critical intersections are compromised

The respective formulations, parameters, and interpretations used in this study are summarized in Table 2.1

Tabela 2.1 - Summary of Network Metrics Used in the Study

Metric	**Equation**	**Symbols**	**Interpretation**
Betweenness Centrality	$C_B(n) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(n)}{\sigma_{st}}$	σ_{st}: shortest paths between s and t; $\sigma_{st}(n)$: paths via n.	Identifies nodes acting as **"bridges"**. In the hydrological network, this highlights vulnerable confluence points; in streets, arterial intersections; in mobility, key hubs controlling flows.
Global Efficiency	$G_E = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}}$	G_E: Global efficiency; N: Total nodes; d_{ij}: Weighted shortest path distance between nodes i and j.	Measures the overall efficiency of flow exchange across the entire network. A **higher global efficiency** indicates better overall connectivity and robustness.
Node Vulnerability	$V_i = \frac{E_{\text{orig}} - E_{\text{without } i}}{E_{\text{orig}}}$	V_i: Vulnerability of node i; E_{orig}: efficiency of the full network.	Represents the relative functional importance of a node. A **higher** V_i indicates that the removal of the node results in a larger decrease in global efficiency, signifying greater structural criticality.

2.6 Analytical Techniques for Comparison and Statistical Validation

To visually compare the distribution of betweenness centrality and vulnerability across different node groups (all nodes, flood-affected nodes, and randomly selected non-flood nodes), Kernel Density Estimate (KDE) and Empirical Cumulative Distribution Function (ECDF) plots were utilized. These plots provide insights into the shape, spread, and shifts in the distributions. For vulnerability it allowed an intuitive understanding of how the removal of nodes from each group impacts the network's global efficiency.

- **KDE Plots**: The Kernel Density Estimate plots were generated to show the probability density of betweenness centrality values for each group.
- **ECDF Plots**: The Empirical Cumulative Distribution Function plots provided a cumulative view of the centrality values.

To statistically validate the observed visual differences in betweenness centrality and vulnerability distributions between flood-affected nodes and random non-flood nodes, the Kolmogorov–Smirnov (KS) test and the Mann–Whitney U test was utilized.

- **Kolmogorov–Smirnov (KS) Test**: The KS test was used to determine if the two underlying probability distributions (between flood-affected and random non-flood nodes) are significantly different. The test statistics quantify the maximum difference between the empirical cumulative distribution functions of the two samples.
- **Mann–Whitney U Test**: The Mann–Whitney U test, also known as the Mann–Whitney–Wilcoxon test, is a non-parametric test that assesses whether two independent samples were drawn from populations with the same distribution. It is particularly useful for comparing whether one group tends to have observations with larger (or smaller) values than the other group.

To complement the significance tests and provide a substantive interpretation of the magnitude of group differences, the median values of betweenness centrality and vulnerability are reported for each group, expressed as percentage differences and in terms of relative network efficiency loss.

By combining visual distribution analysis with non-parametric statistical validation, the study ensured that observed variations in betweenness centrality and vulnerability were not only descriptive but also empirically significant, thereby strengthening the reliability of subsequent interpretations and conclusions.

3 RESULT AND DISCUSSIONS

3.1 Betweenness Centrality

Centrality computation across all nodes produced a highly skewed distribution as seen in Figure 3.1a, where only a few intersections carried most of the thorough traffic potential. When centrality was mapped (Figure 3.1b), it revealed that high-centrality nodes concentrated along the main arterial corridors, notably near the central Tamanduateí Avenue. Low-centrality nodes dominated peripheral neighborhoods, especially within recurrent flood areas.

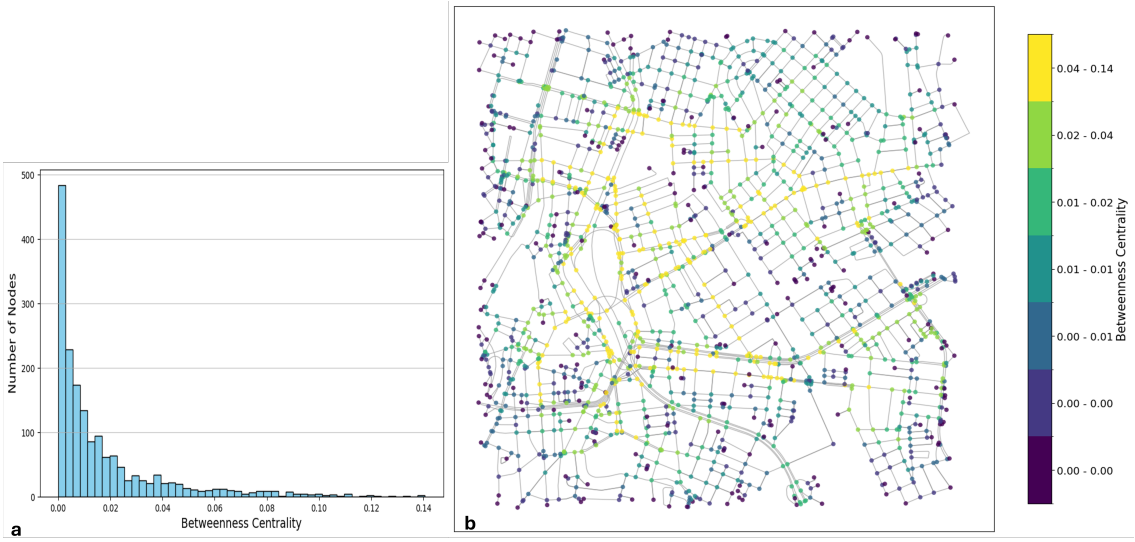

Figura 3.1 - Centrality Distribution Analysis for all Nodes (a) statistical distribution and its (b) geospatial distribution.

This implies that floods disproportionately affect critical connectors. Even though only a fraction of the nodes was inundated, these included some with above-average centrality, meaning small-scale flooding can cause citywide mobility disruptions.

Both the histogram and KDE plot show a distribution that is heavily right-skewed (Figure 3.2). This means that most flood-affected nodes have relatively low betweenness centrality, but there's a long tail extending to the right, indicating a significant number of flood-affected nodes with higher betweenness centrality values, thereby acting as critical connections in the network.

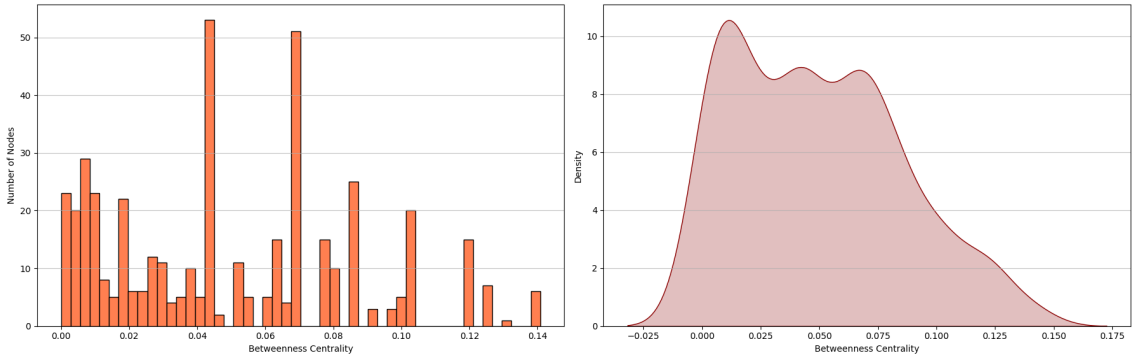

Figura 3.2 - Centrality Distribution Analysis for Flood-affected Nodes (a) statistical distribution and its (b) KDE Plot

The strong skew towards lower betweenness centrality suggests that many flood events occur at nodes that are not necessarily the most central in terms of network flow. These might be localized flood points that disrupt smaller segments of the network rather than major arteries.

Despite the skew, the presence of a tail extending to higher centrality values is crucial. This signifies that a notable portion of flood-affected nodes do coincide with highly central locations. These are the nodes whose disruption would have a significant impact on the overall network's efficiency and connectivity.

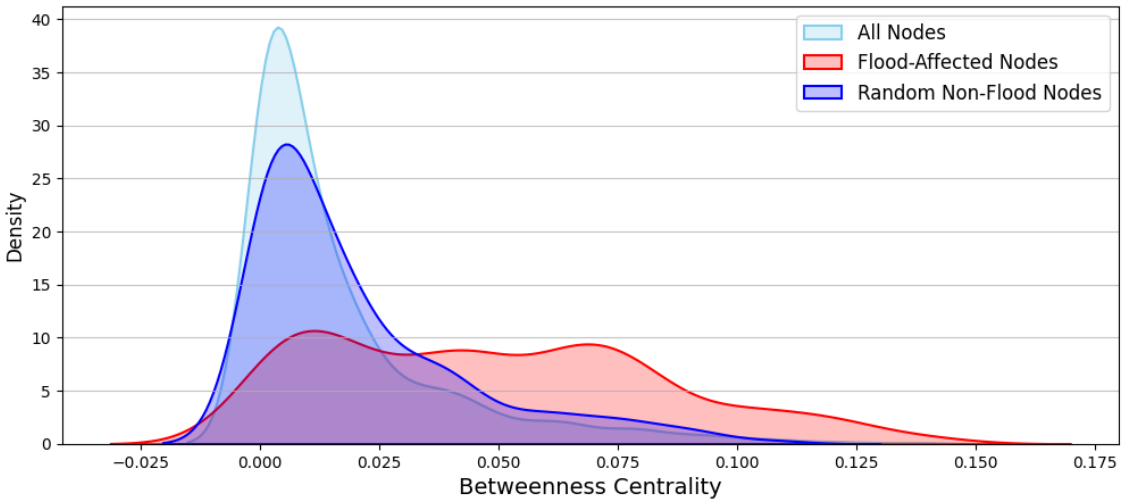

Figura 3.3 - KDE plot comparing all node categories.

When looking at the comparison plot in Figure 3.3, the most striking insight comes from comparing the red curve ('Flood-Affected Nodes') with the green curve ('All Nodes'). The red curve is visibly shifted to the right, indicating that flood-affected nodes, on average, possess higher betweenness centrality. This reinforces the idea that flood events tend to occur in more topologically important locations within the network than a randomly chosen node from the entire graph.

The comparison between the red curve and the blue curve ('Random Non-Flood Nodes') is crucial as well. While both the red and blue curves show a higher average centrality than the 'All Nodes' curve, the red curve still tends to have a slightly higher density at moderately higher betweenness centrality values compared to the blue curve. This suggests that even within a localized area, flood events are not uniformly distributed but are somewhat biased towards nodes with higher betweenness centrality.

This plot strongly suggests that areas experiencing flooding are not just random spots on the map, but they often coincide with critical junctions and segments that are vital for network flow. The fact that flood-affected nodes exhibit higher centrality even when compared to other nodes in the same sample area highlights a systemic vulnerability where environmental hazards intersect with network criticality.

The implication for urban planning and disaster management is clear: efforts to build flood resilience should prioritize nodes with high betweenness centrality that are also prone to flooding. Investing in robust infrastructure, improved drainage, or alternative route planning specifically for these high-centrality, flood-affected nodes would yield the greatest benefit in maintaining network functionality and minimizing disruption during flood events.

3.2 Vulnerability

At first glance (Figure 3.4), the two curves for vulnerability distributions appear somewhat similar, both peaking at lower vulnerability values and having tails extending towards higher vulnerability. This suggests that both flood-affected and random non-flood nodes can exhibit a range of vulnerabilities, and many nodes (regardless of flood status) may not be exceptionally critical to the network's overall efficiency.

Although the overall shapes are similar, upon closer inspection, the red curve ('Flood-Affected Nodes') often shows a slightly higher density or extends further

into the higher vulnerability range compared to the blue curve. This indicates that while not every flood-affected node is highly vulnerable, there's a tendency for a proportionally higher number of highly vulnerable nodes to be among the flood-affected ones.

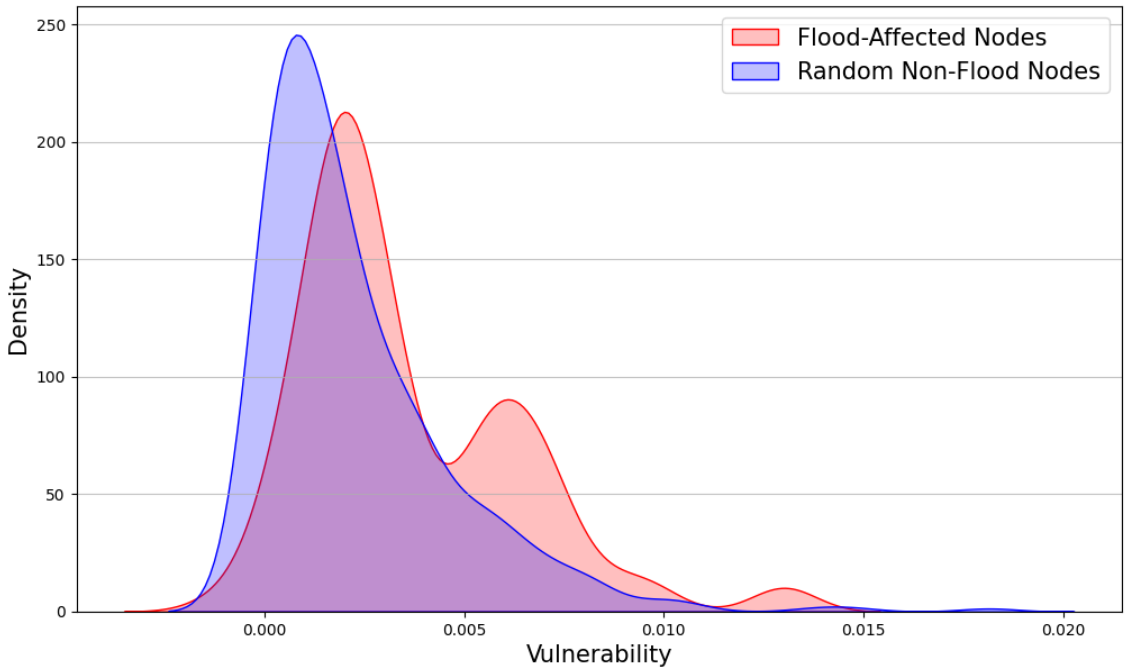

Figura 3.4 - KDE Plot Comparing Node Vulnerabilities

This finding reinforces the earlier observation from betweenness centrality: nodes susceptible to flooding are not random. They are more likely to be locations whose failure (due to flooding, here simulated as removal) would significantly impair the network's efficiency.

In essence, the vulnerability plot, especially when viewed alongside the betweenness centrality distributions, strengthens the argument that flood events in this urban network are not just an inconvenience but pose a significant threat to the network's structural integrity and operational efficiency, particularly because they tend to impact critical nodes more frequently.

3.3 Empirical Cumulative Distribution Function (ECDF)

In Figure 3.5, the blue curve ('Flood-affected') is generally to the right of the orange curve ('Not affected') across a significant portion of the plot. This means that for any

given percentile (y-value), the betweenness centrality value for flood-affected nodes is higher than that of non-flood nodes. This plot provides strong visual evidence that flood-affected nodes tend to have higher betweenness centrality compared to randomly selected non-flood nodes. This confirms the findings from the KDE plots, indicating that nodes susceptible to flooding are, on average, more central to the network's connectivity.

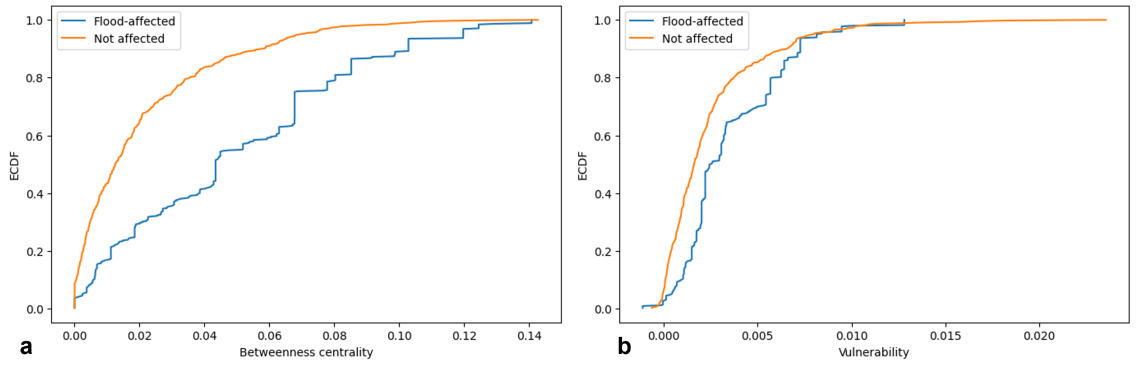

Figura 3.5 - ECDF Plot of (a) Betweenness Centrality and (b) Vulnerability

3.4 Statistical Validation

The Kolmogorov–Smirnov and Mann–Whitney U tests for the betweenness centrality measures yielded extremely small p-values (Table 3.1). This indicates that the observed differences in betweenness centrality distributions between flood-affected nodes and random non-flood nodes are highly statistically significant.

Similar to the values for the betweenness centrality, the test for vulnerability yielded extremely low p-values (Table 3.1), indicating that the observed differences in vulnerability distributions between flood-affected nodes and random non-flood nodes are highly statistically significant and implies that flood-affected nodes tend to have significantly higher vulnerability values than random non-flood nodes.

Tabela 3.1 - Statistical validation and effect size measures for metric distributions (Flood-Affected vs. Random Non-Flood Nodes).

Metric	**Med. (Flood)**	**Med. (Control)**	**Statistical Test**	***p*-value**	**Effect Size**
Betweenness Centrality	0.0276	0.0108	Kolmogorov–Smirnov Mann–Whitney U	9.037×10^{-37} 1.706×10^{-37}	156% higher
Node Vulnerability	0.0024	0.0016	Kolmogorov–Smirnov Mann–Whitney U	1.79×10^{-18} 7.40×10^{-17}	49% greater

In addition to being statistically significant, the magnitude of difference between groups indicates that these effects are substantial. The median betweenness centrality of flooded nodes was 0.0276, which is about a 156% increase over the non-flooded node's median of 0.0108. With respect to vulnerability, the flooded nodes had a median value of 0.0024 and the non-flooded nodes had a median value of 0.0016, corresponding to a median network efficiency decrease upon node removal of 0.24% (flooded nodes) and 0.16% (non-flooded nodes), respectively. Therefore, removing a flooded node from the global network would result in an approximate relative loss of global network efficiency of 49% greater than removing a non-flooded node. This indicates that the structural effects of disruptive events related to floods on urban mobility are not only statistically significant but are also substantively significant.

These findings suggest a structural overlap between flood exposure and network criticality, meaning that environmental hazards tend to impact nodes that are essential for maintaining urban mobility.

4 FINAL REMARKS AND FUTURE WORK

The systemic vulnerability of the urban network is primarily revealed by two key aspects: the distribution of betweenness centrality among flood-affected nodes and the resulting impact on global efficiency (vulnerability). The KDE plots and ECDF comparisons clearly illustrate that flood-affected nodes are not merely random points of failure but are structurally important. The Kolmogorov–Smirnov test and Mann–Whitney U test both confirm statistically significant differences in the betweenness centrality distributions between flood-affected and non-flood nodes ($p < 10^{-37}$), with flood-affected nodes exhibiting a median betweenness centrality 156% higher than non-flood nodes, highlighting a systematic targeting of central infrastructure by flood events.

The vulnerability analysis, which quantifies the reduction in global efficiency upon node removal, corroborates this systemic issue. Flood-affected nodes showed a median vulnerability 49% greater than non-flood nodes, corresponding to a 0.24% versus 0.16% relative reduction in global network efficiency upon removal ($p < 10^{-18}$). While there's a strong correlation between betweenness centrality and vulnerability in both flood-affected and non-flood nodes (Spearman correlation coefficients around 0.72), the locations of high vulnerability coincide with areas prone to flooding. This suggests that the urban design itself, combined with hydrological factors, creates a feedback loop where critical infrastructure is repeatedly exposed to hazards, leading to compounded negative impacts.

This insight can inform more effective urban planning and disaster management strategies. Instead of treating all nodes equally, resources for flood mitigation, infrastructure reinforcement, or emergency response can be strategically allocated to flood-prone areas that also contain highly vulnerable network nodes. Prioritizing these areas could minimize the cascading effects of flooding on the transportation network.

Understanding the spatial overlap between flood risk and network vulnerability allows for the development of more resilient urban systems. This could involve rerouting critical pathways, build redundant infrastructure, or implement smart traffic management systems that can adapt to the sudden loss of key nodes during a flood event.

Future work will extend the analytical framework by incorporating measured river-level data from hydrological stations located near identified flood points. Integrating

these temporal water-level records will enable the transformation of currently static flood-point data into dynamic, time-dependent flood polygons, allowing for the continuous representation of inundation extent and intensity. By correlating river-stage fluctuations with spatial variations in flood coverage, the analysis will facilitate a comparative visualization of vulnerability distributions under multiple hydrological conditions. This, in turn, will help identify critical threshold levels at which the urban transportation network begins to experience accelerated fragmentation or loss of efficiency.

Furthermore, the integration of hydrological and urban network data will support the development of a unified spatiotemporal vulnerability model, bridging the gap between flood dynamics and infrastructure performance. This methodological advancement will shift the study from a static comparison of historical flood impacts to a dynamic resilience assessment, enhancing its predictive capacity and practical relevance for real-time flood risk management, urban infrastructure planning, and adaptive policy formulation in flood-prone metropolitan regions.

REFERÊNCIAS BIBLIOGRÁFICAS

AB'SÁBER, A. N. **Os domínios de natureza no Brasil: potencialidades paisagísticas**. [S.l.]: Ateliê editorial, 2003. 5

AHERN, J. From fail-safe to safe-to-fail: Sustainability and resilience in the new urban world. **Landscape and Urban Planning**, v. 100, n. 4, p. 341–343, 2011. 1

BI, S.; YUAN, X.; HU, S.; LI, K.; NI, W.; HOSSAIN, E.; WANG, X. Resilience and failure analysis in next-generation communication networks: A contemporary survey. **IEEE Transactions on Network Science and Engineering**, 2025. 1

BOCCALETTI, S.; LATORA, V.; MORENO, Y.; CHAVEZ, M.; HWANG, D.-U. Complex networks: Structure and dynamics. **Physics Reports**, v. 424, n. 4–5, p. 175–308, 2006. 1

BOEING, G. Osmnx: A python package to work with graph-theoretic openstreetmap street networks. **The Journal of Open Source Software**, v. 2, n. 12, p. 215, 2017. Available from: <<https://doi.org/10.21105/joss.00215>>. 6

CRUCITTI, P.; LATORA, V.; PORTA, S. Centrality measures in spatial networks of urban streets. **Physical Review E—Statistical, Nonlinear, and Soft Matter Physics**, v. 73, n. 3, p. 36125, 2006. 1, 8, 9

DERRIBLE, S.; KENNEDY, C. Applications of graph theory and network science to transit network design. **Transport Reviews**, v. 31, n. 4, p. 495–519, 2011. 1

KERMANSHAH, A.; DERRIBLE, S. Robustness of road systems to extreme flooding: using elements of gis, travel demand, and network science. **Natural Hazards**, v. 86, n. 1, p. 151–164, 2017. 2

LATORA, V.; MARCHIORI, M. Efficient behavior of small-world networks. **Physical Review Letters**, v. 87, n. 19, p. 198701, 2001. 1, 8

LV, B.; PENG, Y.; DU, C.; TIAN, Y.; WU, J. Review of brillouin distributed sensing for structural monitoring in transportation infrastructure. **Infrastructures**, v. 10, n. 6, p. 148, 2025. 2

MARENGO, J. A.; CAMARINHA, P. I.; ALVES, L. M.; DINIZ, F.; BETTS, R. A. Extreme rainfall and hydro-geo-meteorological disaster risk in 1.5, 2.0, and 4.0°c global warming scenarios: An analysis for brazil. **Frontiers in Climate**, v. 3, 2021. Available from: <<https://doi.org/10.3389/fclim.2021.610433>>. 5

MEEROW, S.; NEWELL, J. P. Urban resilience for whom, what, when, where, and why? **Urban Geography**, v. 40, n. 3, p. 309–329, 2019. 1

MERZ, B.; HALL, J.; DISSE, M.; SCHUMANN, A. Fluvial flood risk management in a changing world. **Natural Hazards and Earth System Sciences**, v. 10, n. 3, p. 509–527, 2010. 1

PORTA, S.; CRUCITTI, P.; LATORA, V. The network analysis of urban streets: A dual approach. **Physica A: Statistical Mechanics and Its Applications**, v. 369, n. 2, p. 853–866, 2006. 1

SABERI, M.; MAHMASSANI, H. S.; BROCKMANN, D.; HOSSEINI, A. A complex network perspective for characterizing urban travel demand patterns: graph theoretical analysis of large-scale origin–destination demand networks. **Transportation**, v. 44, n. 6, p. 1383–1402, 2017. 2

ZHAO, B.; TANG, Y.; WANG, C.; ZHANG, S.; SOGA, K. Evaluating the flooding level impacts on urban metro networks and travel demand: behavioral analyses, agent-based simulation, and large-scale case study. **Resilient Cities and Structures**, v. 1, n. 3, p. 12–23, 2022. 1

PUBLICAÇÕES TÉCNICO-CIENTÍFICAS EDITADAS PELO INPE

Teses e Dissertações (TDI)

Teses e Dissertações apresentadas nos Cursos de Pós-Graduação do INPE.

Notas Técnico-Científicas (NTC)

Incluem resultados preliminares de pesquisa, descrição de equipamentos, descrição e ou documentação de programas de computador, descrição de sistemas e experimentos, apresentação de testes, dados, atlas, e documentação de projetos de engenharia.

Propostas e Relatórios de Projetos (PRP)

São propostas de projetos técnico-científicos e relatórios de acompanhamento de projetos, atividades e convênios.

Publicações Seriadas

São os seriados técnico-científicos: boletins, periódicos, anuários e anais de eventos (simpósios e congressos). Constam destas publicações o Internacional Standard Serial Number (ISSN), que é um código único e definitivo para identificação de títulos de seriados.

Pré-publicações (PRE)

Todos os artigos publicados em periódicos, anais e como capítulos de livros.

Manuais Técnicos (MAN)

São publicações de caráter técnico que incluem normas, procedimentos, instruções e orientações.

Relatórios de Pesquisa (RPQ)

Reportam resultados ou progressos de pesquisas tanto de natureza técnica quanto científica, cujo nível seja compatível com o de uma publicação em periódico nacional ou internacional.

Publicações Didáticas (PUD)

Incluem apostilas, notas de aula e manuais didáticos.

Programas de Computador (PDC)

São a seqüência de instruções ou códigos, expressos em uma linguagem de programação compilada ou interpretada, a ser executada por um computador para alcançar um determinado objetivo. Aceitam-se tanto programas fonte quanto os executáveis.